

```
#include<stdio.h>
#define STACKSIZE 4
#define TRUE 1
#define FALSE 0
```

```
struct stack
{
int top;
    struct student
    {
        int regno;
        char name[10];
        int marks;
    }st[STACKSIZE];
};
```

```
struct stack s;
```

```
void push();
void pop();
int empty();
void main()
{
    char ch;
    int c,x,i;
    s.top=-1;
    do
    {
        printf("MENU\n");
        printf("1. PUSH \n 2. POP\n 3. Display\n ");
        scanf("%d",&c);
        switch(c)
        {
```

```

        case 1: push();
                break;
        case 2: pop();
                break;
        case 3: printf("the elements are\n");
                for(i=s.top;i>=0;i--)
                {
                        printf("the regno : %d\n", s.st[i].regno);
                        printf("the name : %s\n",s.st[i].name);
                        printf("the marks : %d\n", s.st[i].marks);
                }
                break;
    }
    printf("do you wish to continue\n");
    scanf(" %c",&ch);
    }while(ch!='n');
}

```

```

void push()
{
    if(s.top==STACKSIZE-1)
    {
        printf("stack overflow\n");
    }
    else
    {
        s.top++;
        printf("enter the regno\n");
        scanf("%d",&s.st[s.top].regno);
        printf("enter the name\n");
        scanf("%s", s.st[s.top].name);
        printf("enter the marks\n");
        scanf("%d",&s.st[s.top].marks);
    }
}

```

```
    }  
}
```

```
void pop()
```

```
{  
    if(empty())  
    {  
        printf("stack underflow\n");  
    }  
    else  
    {  
        printf("The deleted elements are\n ");  
        printf("the regno : %d\n", s.st[s.top].regno);  
        printf("the name : %s\n",s.st[s.top].name);  
        printf("the marks : %d\n", s.st[s.top].marks);  
        s.top--;  
    }  
}
```

```
int empty()
```

```
{  
    if (s.top==-1)  
        return TRUE;  
    else  
        return FALSE;  
}
```